

09 · Advanced interp pipelines

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July 1, 2026

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Compositions that combine capture, branching, steering, patching, and analyze into real experiments. These lean on the fact that `branch()` is free (shared weights) and every interp tool rides the same generation path.

1 · Difference-of-means direction → causal steering test

Find a direction that separates two conditions, then verify it's *causal* by steering with it.

```
import torch
from interlens import Conversation, SteeringSpec

conv = Conversation.from_models(("Qwen/Qwen2.5-3B-Instruct", "Qwen/
    Qwen2.5-3B-Instruct"), names=("a", "b"))
conv.transcript.append("a", "Tell me your honest opinion of
    pineapple pizza.")

LAYER = 14
def mean_answer_act(prompt):
    b = conv.branch()
    with b.capture(sites=["residual"], layers=[LAYER]) as cache:
        b.sample("b", prompt) # ephemeral;
        nothing committed
    rec = cache.query(participant="b", layer=LAYER)[0]
    lo, hi = rec.phases["answer"]
    return rec.tensor[lo:hi].mean(0)

pos = mean_answer_act("Respond with maximum enthusiasm.")
neg = mean_answer_act("Respond with cold indifference.")
```

```

direction = (pos - neg); direction /= direction.norm()
# Causal check: does adding the direction shift behavior?
for coef in (0.0, 4.0, 8.0):
    steer = SteeringSpec(direction=direction, layers=(LAYER,),
        coef=coef, mode="add")
    print(coef, "→", conv.sample("b", "Your opinion?",
        steering=steer).content[:100])

```

2 · Activation patching across clean/corrupt branches (causal tracing)

Localize *where* a piece of context matters by transplanting clean activations into a corrupted run, layer by layer.

```

from interlens import Patch
base = conv.branch()
base.transcript.append("a", "The secret code is FALCON. What is the
    secret code?")

# clean: capture residuals everywhere we might patch
with base.capture(sites=["residual"], layers=list(range(0, 28, 4)))
    as clean_cache:
        base.step(base.participant("b"))

# corrupt: overwrite the code, then patch one layer's residual back
# from the clean run
def patched_answer(layer, positions):
    donor = clean_cache.at(participant="b", layer=layer)
    patch = Patch(activations=donor[list(positions)], layer=layer,
        positions=positions)
    c = conv.branch()
    c.transcript.append("a", "The secret code is SPARROW. What is
        the secret code?")
    return c.sample("b", patch=patch).content

for L in range(0, 28, 4):
    print(L, "→", patched_answer(L, positions=(6, 7, 8))) # which
        layer restores "FALCON"?

```

3 · Probe-in-the-loop analyze at scale

Collect an activation-based measurement over many rollouts. The probe read happens **inside the worker** while models are resident; only the scalar crosses back.

```
from interlens import Conversation, AutoModelParticipant,
    register_analyzer
import torch

PROBE = torch.load("stance_probe_layer14.pt")  # a [d_model]
        direction you fit earlier

def project_stance(conv):
    with conv.capture(sites=["residual"], layers=[14]) as cache:
        conv.sample("a", "State your position plainly.")
    rec = cache.query(participant="a", layer=14)[0]
    lo, hi = rec.phases["answer"]
    v = rec.tensor[lo:hi].mean(0).float()
    return {"stance_proj": float(v @ PROBE.to(v.device) /
        PROBE.norm())}

register_analyzer("project_stance", project_stance)

conv = Conversation(
    participants=[AutoModelParticipant.from_pretrained("Qwen/
        Qwen2.5-3B-Instruct", name="a"),
        AutoModelParticipant.from_pretrained("Qwen/
        Qwen2.5-3B-Instruct", name="b")],
    shared_context="Debate: should we colonize Mars?",
).turns(6).analyzer("project_stance")
report = conv.rollout(n=128, out_dir="runs/mars")
projs = [r.analysis["stance_proj"] for r in report.results.values()]
    if r.error is None]
print("mean stance projection:", sum(projs) / len(projs))
```

4 · Steering sweep with reproducible turns

Because sample is ephemeral and branch is free, a full coefficient × layer sweep is a couple of loops — pin seed so each cell is comparable.

```
conv.participant("b").seed = 0  # fix RNG so
    only the intervention varies
results = {}
```

```
for layer in (8, 12, 16, 20):  
    for coef in (2.0, 4.0, 8.0):  
        steer = SteeringSpec(direction=direction, layers=(layer,),  
                             coef=coef, mode="add")  
        results[(layer, coef)] = conv.sample("b", "Your opinion?",  
                                             steering=steer).content
```

Gotchas worth remembering

- Interp on an APIParticipant **raises** — use a local ModelParticipant.
- Steering/patch **disable KV reuse** automatically (the cached KV predates the intervention).
- Keep capture **narrow** (few layers, offload="cpu") — all-layers × all-tokens × many-rollouts OOMs.
- branch() shares participant *objects*; mutating a participant's .temperature/.seed on a branch affects the shared object — set them per use or re-set before each run.

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